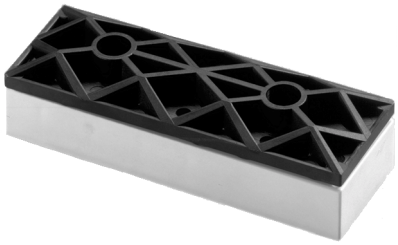
SKiiP 83 ANB 15 T4

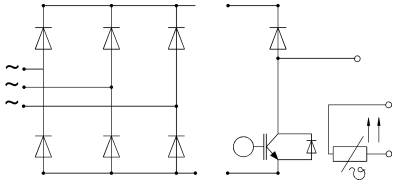
Absolute Maximum Ratings			
Symbol	**Conditions** [1]	**Values**	**Units**
Bridge Rectifier			
V_{RRM}		1500	V
I_D	$T_{heatsink}$ = 80 °C	125	A
I_{FSM}/I_{TSM}	t_p = 10 ms; sin. 180 °C, T_j = 25 °C	1000	A
I^2t	t_p = 10 ms; sin. 180 °C, T_j = 25 °C	5000	A^2s
IGBT Chopper			
V_{CES}		1200	V
V_{GES}		± 20	V
I_C	$T_{heatsink}$ = 25 / 80 °C	125 / 85	A
I_{CM}	t_p < 1 ms; $T_{heatsink}$ = 25 / 80 °C	250 / 170	A
Freewheeling Diode [2]			
V_{RRM}		1200	V
I_F	$T_{heatsink}$ = 25 / 80 °C	80 / 55	A
I_{FM}	t_p < 1 ms; $T_{heatsink}$ = 25 / 80 °C	160 / 110	A
T_j	Diode & IGBT	– 40 ... + 150	°C
T_{stg}		– 40 ... + 125	°C
V_{isol}	AC, 1 min.	2500	V

Characteristics					
Symbol	**Conditions** [1]	**min.**	**typ.**	**max.**	**Units**
Diode - Rectifier					
V_F	I_F = 100 A T_j = 125 °C	–	1,15	–	V
V_{TO}	T_j = 125 °C	–	0,8	–	V
r_T	T_j = 125 °C	–	3,5	–	mΩ
R_{thjh}	per diode	–	–	0,7	K/W
IGBT - Chopper					
V_{CEsat}	I_C = 100 A T_j = 25 (125) °C	–	2,5(3,1)	3,0(3,7)	V
$t_{d(on)}$	V_{CC} = 600 V; V_{GE} = ± 15 V	–	50	100	ns
t_r	I_C = 100 A; T_j = 125 °C	–	55	110	ns
$t_{d(off)}$	R_{gon} = R_{goff} = 11 Ω	–	400	600	ns
t_f	inductive load	–	70	100	ns
E_{on} + E_{off}		–	27	–	mJ
C_{ies}	V_{CE} = 25 V; V_{GE} = 0 V, 1 MHz	–	6,6	–	nF
R_{thjh}	per IGBT	–	–	0,25	K/W

MiniSKiiP 8
SEMIKRON integrated intelligent Power
SKiiP 83 ANB 15 T4
3-phase bridge rectifier + IGBT braking chopper

Case M8a





UL recognized file no. E63532

- specification of temperature sensor see part A of data book ´99
- common characteristics see page B 16 – 4 of data book ´99

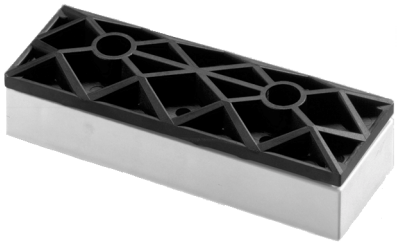
[1] $T_{heatsink}$ = 25 °C, unless otherwise specified
[2] CAL = Controlled Axial Lifetime Technology (soft and fast recovery)

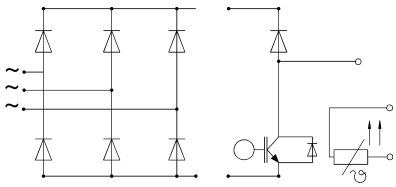
MiniSKiiP 8
SEMIKRON integrated intelligent Power

SKiiP 83 ANB 15 T4

3-phase bridge rectifier + IGBT braking chopper

Case M8a





SKiiP 83 ANB 15 T4

Characteristics					
Symbol	**Conditions** [1]	**min.**	**typ.**	**max.**	**Units**
Diode [2] - Freewheeling					
$V_F = V_{EC}$	I_F = 75 A T_j = 25 (125) °C	–	2,0(1,8)	2,5(2,3)	V
V_{TO}	T_j = 125 °C	–	1,0	1,2	V
r_T	T_j = 125 °C	–	11	15	mΩ
I_{RRM}	I_F = 75 A; V_R = – 600 V	–	45	–	A
Q_{rr}	di_F/dt = – 800 A/µs	–	11	–	µC
E_{off}	V_{GE} = 0 V, T_j = 125 °C	–	3,0	–	mJ
R_{thjh}	per diode	–	–	0,8	K/W
Diode [2] - Antiparallel					
$V_F = V_{EC}$	I_F = 15 A T_j = 25 (125) °C	–	2,0(1,8)	2,5(2,3)	V
V_{TO}	T_j = 125 °C	–	1,0	1,2	V
r_T	T_j = 125 °C	–	53	73	mΩ
I_{RRM}	I_F = 15 A; V_R = – 600 V	–	16	–	A
Q_{rr}	di_F/dt = – 400 A/µs	–	2,7	–	µC
E_{off}	V_{GE} = 0 V, T_j = 125 °C	–	0,6	–	mJ
R_{thjh}	per diode	–	–	1,7	K/W
Temperature Sensor					
R_{TS}	T = 25 / 100 °C		1000 / 1670		Ω
Mechanical Data					
M_1	case to heatsink, SI Units	2,5	–	3,5	Nm
Case	mechanical outline see pages B 16 –13 and B 16 – 14		M8a		

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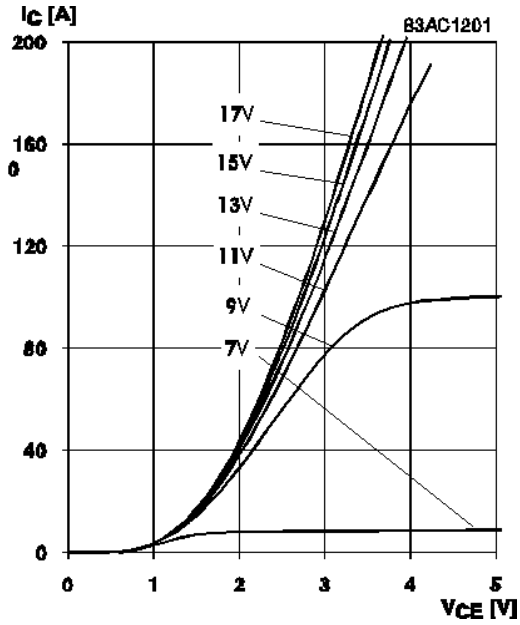


Fig. 1 Typ. output characteristic, t_p = 80 µs; 25 °C

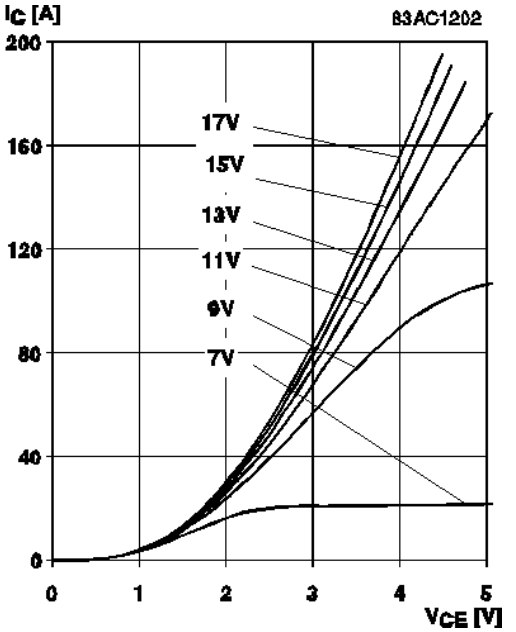
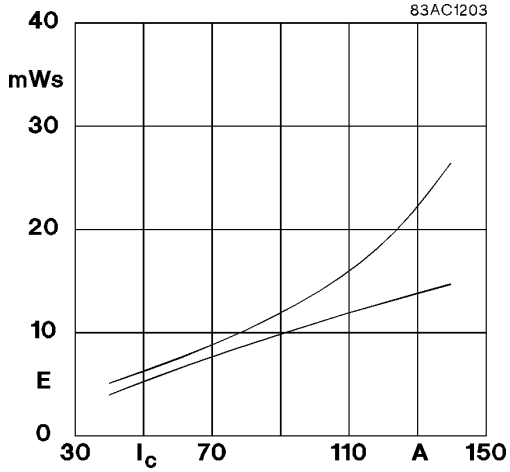
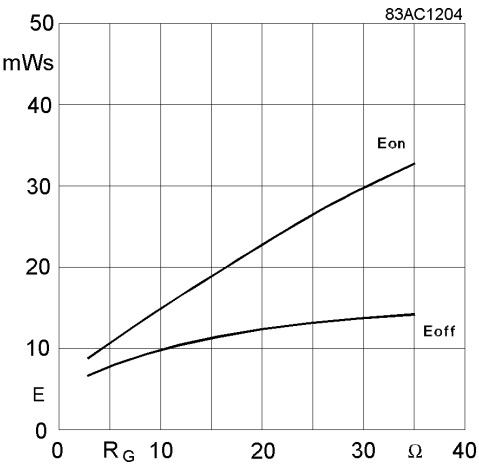


Fig. 2 Typ. output characteristic, t_p = 80 µs; 125 °C



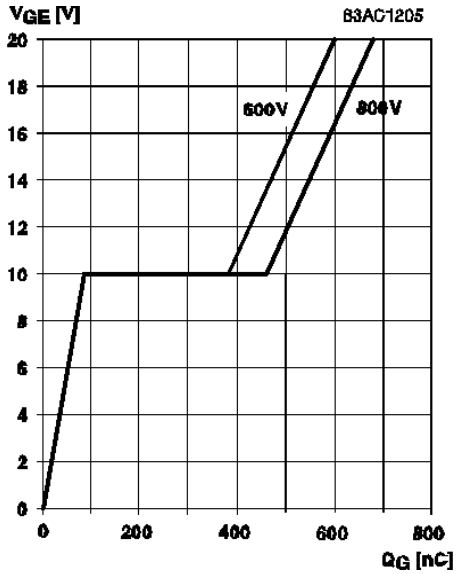
T_j = 125 °C
V_{CE} = 600 V
V_{GE} = ± 15 V
R_G = 11 Ω

Fig. 3 Turn-on /-off energy = f (I_C)



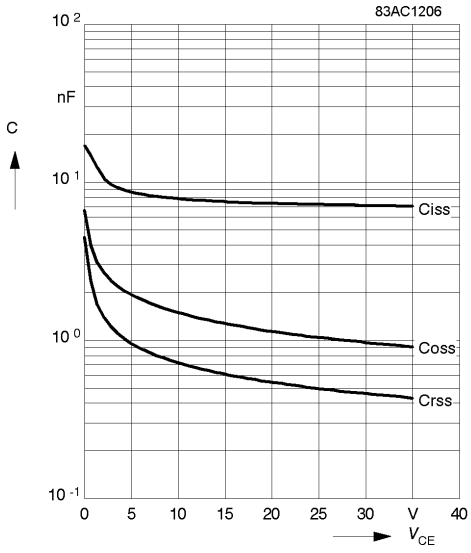
T_j = 125 °C
V_{CE} = 600 V
V_{GE} = ± 15 V
I_C = 100 A

Fig. 4 Turn-on /-off energy = f (R_G)



I_{Cpuls} = 100 A

Fig. 5 Typ. gate charge characteristic



V_{GE} = 0 V
f = 1 MHz

Fig. 6 Typ. capacitances vs. V_{CE}